
Education

University at Buffalo-Roswell Park Comprehensive Cancer Center | Buffalo, NY, USA

Ph.D. Cancer Sciences

2019 - 2025

- ★ Thesis focus: Immuno-oncology and targeted therapy

M.S. Cancer Sciences

2017 - 2019

- ★ Thesis focus: DNA damage response to small molecule anti-cancer agents

PSG College of Technology | Coimbatore, India

BTech. Biotechnology

2013 - 2017

- ★ Thesis focus: Molecular mechanisms of cell division

Experience

Independent Researcher

Apr 2025 - present

- ★ [Tracing the cellular and tissue origins of plasma protein biomarkers for early cancer detection by applying machine learning workflows to large single cell datasets.](#)
- ★ [Building a tumor immune-suppression signature predictive of immunotherapy response and pre-metastatic priming.](#)
- ★ [Quantifying BRCA variant frequencies across Indian sub-populations using the GenomeIndia dataset.](#)
- ★ [Developing Medindicator, an app that flags drug-drug interactions using established interaction data.](#)

Senior Research Aide (Pharmacodynamics simulation and modeling)

Enhanced Pharmacodynamics LLC, Buffalo, NY

Jan 2019 - May 2019

- ★ Developed an interactive web app using R Shiny, to predict immune toxicity of an anti-cancer drug candidate, based on existing population pharmacokinetic/pharmacodynamic (PK/PD) models.

Technical skills

Molecular biology: Cloning, CRISPR-Cas9, RT-qPCR, transfection, primer design, maxi and mini preps, gel electrophoresis

Bioinformatics: R, Python, Linux/Command Line, FlowJo, FCSEXPRESS

Protein & Cell biology: Multiparametric flow cytometry (10+ colors), cell culture, immunoblot, proliferation, migration, cell death, reporter, DNA damage detection assays

In vivo Models: Syngeneic and PDX mouse models, drug dosing, blood draw, tissue harvesting & processing

Publications

[Rajan, P., Zollo R., Guo Y., Alruwaili M., Zonneville, J., Lieberman M., Morreale B., Olejniczak S., Barbi J., Abrams, S., & Bakin, A. p38 blockade reverses the immune suppressive tumor microenvironment in metastatic breast cancer. 2025. Under review at NPJ Precision Onc.](#)

[Alruwaili, M., Zonneville, J., Naranjo M., Serio, H., Melendy, T., Straubinger, R., Gillard, B., Foster, B.A., Rajan, P., Attwood K., Chatley, S., Iyer, R., Fountzilias, C., Bakin, A.V. A Synergistic Two-Drug Therapy Specifically Targets a DNA Repair Dysregulation That Occurs in p53-Deficient Colorectal and Pancreatic Cancers. Cell Reports Medicine, 2024, 5\(3\): p. 101434.](#)

[Luzhin A., Rajan P., Safina A., Leonova K., Stablewski A., Wang J., Robinson D., Isaeva N., Kanditze O., Gurova K. \(2023\). Comparison of cell response to chromatin and DNA damage. Nucleic Acids Research, 2023; gkad865.](#)

Awards & Honors

- ★ Completed Claude Science Hackathon (1 of 150 researchers selected worldwide)
- ★ Awarded for Academic Excellence by University at Buffalo
- ★ Supported by \$3000 grant award from University at Buffalo Mark Diamond Research Fund (UB-MDRF)
- ★ Selected for short-term project at Tata Institute of Fundamental Research (TIFR), Mumbai, India